

Graphs

$$G = (V, E)$$



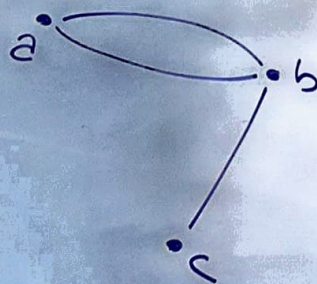
(simple) graph
 $E \subseteq \binom{V}{2}$

✓ simple
(even if we don't say
"simple")

✓ could be embedded
somewhere, but need
not be

multi-graph: allows double edges

(not our usual setting though)



a path in a graph is
a sequence of vertices such
that 2 consecutive verts
are on edge

$$\gamma = [u = u_1, u_2, u_3, \dots, u_k = v]$$

$$(u_i, u_{i+1}) \in E$$

path from u to v

$$\text{length}(\gamma) = l(\gamma) = \begin{matrix} \# \text{ edges} \\ \# \text{ vertices} - 1 \end{matrix}$$

Some sources: sequence of edges
alt. seq. of vert-edge-vert...

a path in a topological space X is a continuous map

$$\gamma: [0,1] \rightarrow X$$

> Note: $[0,1]$ is a subspace of $\mathbb{R}$ (std. topology)

$A \subset [0,1]$ is open

iff $\exists U \overset{\text{open}}{\subseteq} \mathbb{R}$ such that $A = U \cap [0,1]$

Subspace topology

Q: What is the topology of a graph?

(5)

(1) underlying set: V
generators: E
(I've never seen this, but is a topology!)

(2) Discrete $\leftarrow$ sets $V \subseteq E$
underlying space $|G|$

(3) Indiscrete $\leftarrow V \subseteq E$
 $|G|$

(4) Embedded Subspace Topology $|G|$

a) embed into $\mathbb{R}^3$ (always possible)

b) use subspace topology

Thm The embedding doesn't matter!

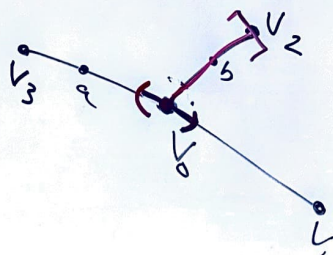
Prof. exercise

paths can be between any 2 points in $|G|$

"underlying space"
"quotient space"

- Quotient Topology
 - consider discrete topo on V
 - $\forall e \in E, e \cong [0, 1]$
 - glue everything together

caveat:
edges are often
thought of as open
intervals.



(5) underlying set: $V \sqcup E$

- all $e \in E$ are open
- $U \subset V \sqcup E$ is open iff $\forall v \in V \cap U$, all edges containing v are also in U .

"paths" restricted to be b/w vertices.

Names

- UP set topology
- upwards closed topu.
- Alexandroff topu.

← example open set



Thm

Consider an embedding of G into $\mathbb{R}^3$. Then, any open set in (4) or in (5) deformation retracts onto a simplicial subcomplex.

path in gen topo space X

$\gamma: [0,1] \rightarrow X$ continuous

$u := \gamma[0]$ the start

$v := \gamma[1]$ the end

We say γ goes from u to v
between u and v .

[Q] Play w/ the topo. def'n of
path + the space (5)

Graphs

Def'n A graph $G=(V,E)$ is connected if
 $\forall v,u \in V, \exists$ a path between
u and v.

Theorem

$G=(V,E)$ is connected iff
 $\nexists$ separator (i.e., $\nexists V_1 \sqcup V_2 = V$
such that $V_i \neq \emptyset$
and there are no edges between
 V_1 and V_2)

Proof: Exercise (using definitions)

Computation: Union-Find (EH 1.2)
CLRS is great!

Topological Spaces

2 no

connectivity:

- ① X topo space is path-connected iff

$\forall x, y \in X, \exists$ a path between x and y .



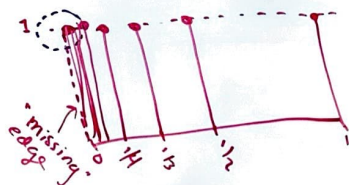
- ② X is connected iff $\nexists$ a separator.

Def'n

U, V open
are a separator
iff $U \cap V = \emptyset$
and $X = U \cup V$

Example of Connected, but not path-connected.

The comb space $\subset \mathbb{R}^2$



$(0,1,0)$
 $U(1/n, 0, 1]$
 $U(0,1)$

Thm ① $\Rightarrow$ ②

Proof: exercise.

UNION-FIND algo

↓ a set (of our potential conn.comp)

$$S = \{V_1, V_2, \dots, V_n\}$$

$E' =$ stack of edges not yet

added

↳ pop = remove top one

↳ as we pop, building $(V_1, \dots)$ into (V, E)
one edge at a time.



NOT conn $\Rightarrow$ must union/merge conn.comp

Are conn $\Rightarrow$

